Project Guidelines

You are expected to follow the best practices of machine learning throughout the project. Make reasonable assumptions when needed. As appropriate to your dataset and task, you should include the following components:

> **Data Preprocessing:** You should carefully explore the dataset and apply appropriate preprocessing steps. Perform data cleaning, including handling missing values, addressing class imbalance, applying normalization or encoding techniques, and conducting feature engineering or selection, train/test split as needed.
>
> **Modeling:** Train and evaluate at least three distinct machine learning models, including both traditional and neural network-based models. If applicable, perform hyperparameter tuning to optimize model performance, model selection, cross-validation, and ensemble models
>
> **Evaluation:** Use appropriate evaluation metrics to examine training and test performance, detecting underfitting or overfitting, and compare the strengths and weaknesses of the models.
>
> **Result Analysis:** Provide a critical analysis of the approaches used, justify which method performed best for the given dataset, and discuss the limitations of your work as well as potential avenues for improvement.

Computational Resources

Make sure you have the computational resources needed for the completion of your project. You may choose to work on your own machine, on cloud services such as Google Colab, or utilize campus resources if you have access. Since this is a class project, you are also allowed to limit the project scope to what is reasonable and feasible within the available resources (downscale models, using a subset of datasets, etc).

GitHub Repository

Every group must create a GitHub repository for the project. All <u>members should contribute to the same repository visibly.</u>

Your repo must contain **at least three top-level directories**:

> **`code/`**: All your scripts, notebooks, and implementations.
> **`resources/`**: Research papers, reference links, notes that you used for the project, and your final presentation slides.
> **`data/`**: Datasets, pre-trained models, etc.

`README.md` Title of the Project, Abstract (a brief paragraph of your project), list of developers(names of all team members), and How to run the project guides (steps, commands, etc.).

A `requirements.txt` file listing all dependencies/libraries required to run your code.

Grading Criteria

Component	**Grading** (I: Individual, G: Group)	**Weightage**
Project Proposal (Group Submission)	G	10%
Project Midterm Check-in 1 (I)	I	5%
Project Midterm Check-in 2	I	5%
Repository Structure & Documentation	G	5%
Implementation, Code Quality, Completeness of the Project	I	20%
Appropriate Use of the AI/ML Algorithm	I	20%
Group Collaboration and Participation (Includes credits for peer evaluation you provide in the presentation of other groups)	I	5%
Presentation 1. Peer Review * 2. Instructor's Review Contribution Form	I	30%

* Each student will evaluate at least 2 other groups based on the provided rubric. To ensure fairness in grading, the instructor holds the right to discard any peer review score that is significantly deviated from the average of the other reviews.